CATERHAM SEVEN

Suspension Setup Guide

Ride height, rake, alignment and damper settings — a working method, and a starting point you can trust.

The figures in this guide are a considered starting point, not a final answer. They will shift with tyre choice, circuit, surface, driver preference and how the car is used. What matters more than any single figure is the method: set the car up the same way every time, change one thing at a time, and keep a record of what you did.

1 Before you start

- A tape measure.
- A block of wood or similar cut to **70mm** for a track setting, or **80mm** for a road setting. This saves a great deal of time and a lot of crawling about later.
- A flat, level surface — a garage floor is fine, a sloping driveway is not.
- Someone to sit in the car, or ballast of a similar weight. A few sacks of spuds will do.

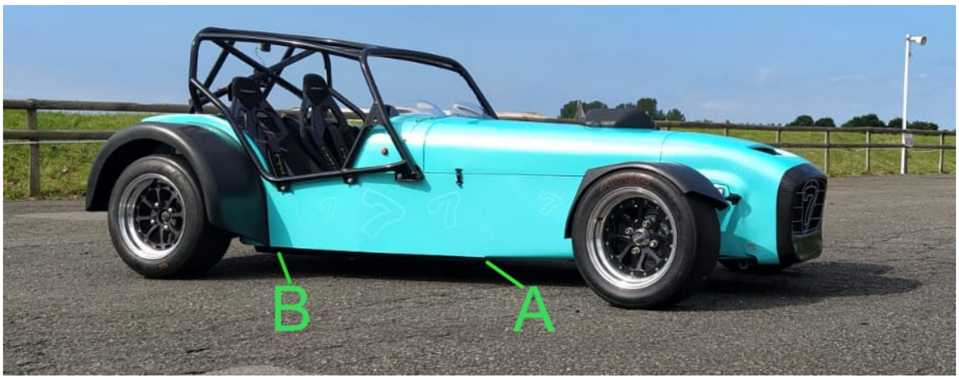
You will have space to adjust the front dampers without removing the wheel. At the rear it is likely you will need to keep taking the wheels off to reach the spring preload.

2 Ride height and rake

Sump clearance is the starting point for the whole setup. Set it first and everything else follows from it. Use **70mm for track use** and **80mm for road use**.

Place the block under the front of the sump pan. Adjust the height of the car at the spring platform: compress the spring further to raise the car, and slacken it off to lower the car. Which way the platform turns to do that depends on how the damper is mounted, so work by what the spring is doing rather than by up or down. Wind it until the block just fits underneath. It is best to jack the car up to do this so the suspension is unloaded. It can be tedious, but stick with it. As a guide, 6mm of thread on the front spring platform gives roughly 10mm of ride height at the front, and 8mm of thread at the rear will lift the rear by around 10mm.

With the sump clearance set, measure from the chassis at **Point A** down to the floor. This is where the lower chassis skin changes from a rounded to a square edge. Use the square edge — it gives a far more accurate reading against the tape.



Point A — lower chassis skin, where the rounded edge becomes square. Point B — lower chassis rail, just in front of the rear wing mount.

Depending on your sump type you should see a figure of **125–145mm** at Point A on a track setting. A road setting, with 80mm under the sump, will read approximately 10mm higher throughout.

Now measure from the lower chassis rail to the floor at **Point B**, just in front of the rear wings where they meet the chassis. This measurement should be **5–10mm greater than Point A**. On higher powered cars, work towards the lower end of that range.

Working through it

This has to be done with you, and any usual passenger, sitting in the car — so you will need help. Expect to go round in circles a few times; three to five cycles is normal in my experience, adding or removing preload on the relevant springs as you go. Bear in mind that adding height at the rear will also add roughly half that value to your front measurement, and the same applies in reverse. Move in small steps and do not go too far in one go.

If you are working alone, a reliable rule of thumb is to set the car to its 10mm of rake, equal on both sides, then add four full turns on the driver's side rear damper adjuster ring to compress that spring. You will not be far off.

Settling the car

Every time you drop the car back onto its dampers, jump up and down on the wishbones and on the boot, and roll the car back and forth — exactly as you would after a session on track. Suspension that has not been settled will give you a measurement you cannot repeat.

> **Why this method works**
> I have been setting my own car up to the rake described here for more than twenty-five years of sprinting, and I consider it a well balanced car. I used this method alone until 2012, when I bought a set of corner weight scales. Measured against the scales, it proved to be remarkably accurate — which is why it is still the method I recommend to customers working at home.

3 Setting up with corner weights

If you are working with corner weight scales there are two real schools of thought, the main one being to balance the car diagonally. I do not personally like a Caterham set up that way.

My method starts in the same place — sump clearance. From there, instead of setting ride heights at the front, I equal the front corner weights with the driver in the car. I then check the height at Point A on both sides, and you will find there may be up to 3mm difference between them. That is fine.

I then ignore the rear corner weights altogether, and simply make sure the rake is the same on both sides.

> **For example**
> 70mm under the sump, and 135kg on each front wheel. At Point A I read 130mm on one side and 133mm on the other. I then set Point B to 138mm and 141mm respectively — the same rake each side. The rear weights are ignored.

From there the car goes out and gets driven. If one front wheel locks before the other under braking, I add a turn of preload to the locking wheel before the next run, and that will probably sort it. If it is still locking, add another turn.

Effectively, I am not interested in what the maths says. All I want is real world balance, and setting the car up this way allows for maximum braking stability.

4 Alignment

The figures below are a good starting point for road use. They will change for you, and they will certainly change with tyre choice and with the move from road to track.

Setting	Front	Rear
Camber	-1.5°	-1.5° (dependent on the de Dion ear)
Caster	As much as is available	—
Toe	0.0mm (parallel)	1 to 1.5mm toe-in each side

Record it — StringAlignPro

Alignment is only useful if you can repeat it. StringAlignPro walks you through a string alignment properly and, more importantly, keeps a record of every measurement you take — so next time you can see exactly what you changed and what it did.

Download: search for StringAlignPro on the App Store or Google Play.

5 Damper settings

Always count back from full hard. Wind the adjuster fully in, then count clicks back out to the figure shown. Check both sides match before you drive the car.

Nitron 1-Way

Use	Front	Rear
Road	-8	-16
Track	-6	-14

3-Way dampers

Damper	Position	Low-speed comp.	High-speed comp.	Rebound
Nitron 3-Way	Front	-6	-8	-8
Nitron 3-Way	Rear	-10	-12	-12
Penske 3-Way	Front	-4	+10	-40
Penske 3-Way	Rear	-12	+8	-40

For any other multi-way damper, specific settings are supplied with the kit and developed around your car and its intended use.

Torque settings — SevenSense

Before anything comes apart, or goes back together, check the figure. SevenSense carries the Caterham torque settings in one place, so you are not guessing on suspension fasteners or hunting through a manual on a garage floor.
Download: search for SevenSense on the App Store or Google Play.

Talk it through

If the car is not doing what you want it to do, the numbers on this sheet are the beginning of the conversation, not the end of it. Suspension is specified around the car, the driver and the use — and a description of what the car is doing under you is often worth more than any measurement.
Too Fast To Race | Simon Rogers | toofasttorace.com

6 Record your setup

Write it down before you drive. A setup you cannot repeat is a setup you cannot develop. The best way to keep this record is in **StringAlignPro**, which will produce a far more professional and complete output than the sheet below. Use this page if you would rather work on paper, or as a quick note in the garage before you transfer it across.

Car / registration	Date
Tyres fitted	Circuit / use

	Front	Rear	Notes
Sump clearance		—	
Point A / Point B			
Camber			
Caster		—	
Toe			
Damper — LSC			
Damper — HSC			
Damper — Rebound			
Spring rate			
Tyre pressures			